

Shuang Chen

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Website

<https://sc2682cornell.github.io>

Research Interests

Computer Architecture and Systems
Hardware-Software Co-Optimization
LLM Infrastructure

EDUCATION

Cornell University, USA.

May 2021

Ph.D. in Computer Engineering

- Committee: José F. Martínez (advisor), Christina Delimitrou, Robbert van Renesse
- Thesis: Improving Resource Efficiency for Colocation of Multiple Latency-Critical Applications in Datacenters

Shanghai Jiao Tong University, China

July 2015

B.S. in Computer Science, ACM Honored Class, Zhiyuan College

- Admitted in Aug 2010, waiving the college entrance exam

PROFESSIONAL EXPERIENCES

Technical Expert at Bytedance Seed, Hangzhou, China.

Apr 2026 ~ Present

LLM Inference Infrastructure team

Technical Expert (Top Minds) at Huawei Cloud, Hangzhou, China.

Aug 2021 ~ Apr 2026

Cloud Software and Hardware Co-Design Lab

Research Intern at Facebook Inc., CA, USA.

Summer 2018

Capacity Engineering and Analysis team, under Dr. Banit Agrawal

Software Engineering Intern at Cavium (now Marvell) Inc., CA, USA.

Summer 2016

Processor Architecture team, under Dr. Srilatha(Bobbie) Manne

PROFESSIONAL SERVICES

- **Program Committee:** MICRO 2026, HPCA 2026, NSDI 2026, YArch 2026/2025/2024/2023, APPT 2025, ChinaSys Spring 2024(PC Co-Chair), ASPLOS 2024/2023
- **External Review Committee,** MICRO 2025, ISCA 2024, ASPLOS 2022, HPCA 2021

PUBLICATIONS

[^] denotes corresponding author.

- **[Preprint]** Weiwei Chen, **Shuang Chen**[^], Lele Li, Qiang Hu, Han Li, Xin Ye, Ming Yan, Zhibin Yu, “ASAP: A Disaggregated and Asynchronous Inference System for MoE Prefill,” on Arxiv.
- **[Preprint]** Ao Xu, Han Zhao, Weihao Cui, Quan Chen, Yukang Chen, Shulai Zhang, **Shuang Chen**, Jiemin Jiang, Zhibin Yu, Minyi Guo, “Harli: SLO-Aware Co-location of LLM Inference and PEFT-based Finetuning on Model-as-a-Service Platforms,” on Arxiv.
- Yuxuan Liu, Yeh-Ching Chung, **Shuang Chen**[^], “Barbell: An Extensible Generator of On-Demand Loads for Interactive Cloud Services,” in *the 17th International Symposium on Advanced Parallel Processing Technology*, July 2026.
- Cunchen Hu, Heyang Huang, Liangliang Xu, Xusheng Chen, Chenxi Wang, Jiang Xu, **Shuang Chen**, Hao Feng, Sa Wang, Yungang Bao, Ninghui Sun, Yizhou Shan, “ShuffleInfer: Disaggregate LLM Inference for

Mixed Downstream Workloads,” in *ACM Transactions on Architecture and Code Optimization (TACO)*, July 2025.

- Yajuan Peng*, **Shuang Chen***(co-first authors), Yi Zhao, Zhibin Yu, “UFO: The Ultimate QoS-Aware Core Management for Virtualized and Oversubscribed Public Clouds,” in *the 21th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, April 2024.
- **Shuang Chen**, Angela Jin, Christina Delimitrou, and José F. Martínez, “ReTail: Opting for Learning Simplicity to Enable QoS-Aware Power Management in the Cloud,” in *the 28th IEEE HPCA International Conference on High Performance Computer Architecture (HPCA)*, Feb 2022.
- **Shuang Chen**, Yi Jiang, Christina Delimitrou, and José F. Martínez, “PIMCloud: QoS-Aware Resource Management of Latency-Critical Applications in Clouds with Processing-in-Memory,” in *the 28th IEEE HPCA International Conference on High Performance Computer Architecture (HPCA)*, Feb 2022.
- **Shuang Chen**, Christina Delimitrou, and José F. Martínez, “PARTIES: QoS-Aware Resource Partitioning for Multiple Interactive Services,” in *the 24th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Apr 2019.
- **Shuang Chen**, Shay Galon, Christina Delimitrou, Srilatha Manne, and José F. Martínez, “Workload Characterization of Interactive Cloud Services on Big and Small Platforms,” in *the 2017 IEEE International Symposium on Workload Characterization (IISWC)*, Oct 2017.
- Xiaodong Wang, **Shuang Chen**, Jeff Setter and José F. Martínez, “SWAP: Effective Fine-grain Management of Shared Last-level Caches with Minimum Hardware Support,” in *the 23rd IEEE HPCA International Conference on High Performance Computer Architecture (HPCA)*, Feb 2017.
- **Shuang Chen**, Shunning Jiang, Bingsheng He and Xueyan Tang, “A Study of Sorting Algorithms Using Approximate Storage,” in *the 2016 ACM SIGMOD International Conference on Management of Data (SIGMOD)*, June 2016.
- Naifeng Jing, Shunning Jiang, **Shuang Chen**, Jinjie Zhang, Li Jiang, Chao Li and Xiaoyao Liang, “Bank Stealing for a Compact and Efficient Register File Architecture in GPGPU,” in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2016.
- Naifeng Jing, **Shuang Chen**, Shunning Jiang, Li Jiang and Xiaoyao Liang, “Bank Stealing for Conflict Mitigation in GPGPU Register File,” in *the 2015 ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, July 2015.

TALKS

QoS-Aware Resource Management in the Cloud: From Academia to Industry

- Invited talk at *IIIS of Tsinghua University*, hosted by Dr. Yi Wu, Feb 21, 2023
- Invited talk at *ICT, Chinese Academy of Sciences*, hosted by Dr. Chenxi Wang, Feb 22, 2023
- Invited talk at *Nankai University*, hosted by Dr. Yusen Li, Feb 27, 2023
- Invited talk at *Tianjin University*, hosted by Dr. Shanjiang Tang, Feb 28, 2023
- Invited talk at *Zhejiang University*, hosted by Dr. Zeke Wang, April 26, 2023

PIMCloud: QoS-Aware Resource Management of Interactive Services in Datacenters with Processing-in-Memory

- Poster session at *SRC TECHCON*, Austin, Sept 9, 2019
- Invited talk at *SRC JUMP e-Workshop*, Oct 15, 2019
- Poster session at *SRC JUMP Center for Research in Intelligent Storage (CRISP) Annual Review*, Charlottesville, USA, Nov 5, 2019
- Invited talk at *CCFSys*, online, Dec 15, 2021
- Conference talk at *the 28th IEEE HPCA International Conference on High Performance Computer Architecture (HPCA)*, online, Apr 6, 2022

ReTail: Opting for Learning Simplicity to Enable QoS-Aware Power Management in the Cloud

- Invited talk at *ChinaSys*, online, Dec 4, 2021
- Conference talk at *the 28th IEEE HPCA International Conference on High Performance Computer Architecture (HPCA)*, online, Apr 4, 2022

PARTIES: QoS-Aware Resource Partitioning for Multiple Interactive Services

- Invited talk at *Microsoft Research*, Redmond, Nov 29, 2017
- Conference talk at *the 24th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Providence, USA, Apr 15, 2019
- Research pitch competition and poster session at *Rising Stars in EECS Workshop*, University of Illinois at Urbana-Champaign, USA, Oct 30, 2019

Workload Characterization of Interactive Cloud Services on Big and Small Platforms

- Conference talk at *the 2017 IEEE International Symposium on Workload Characterization (IISWC)*, Seattle, USA, Oct 2, 2017

A Study of Sorting Algorithms Using Approximate Storage

- Conference talk at *the 2016 ACM SIGMOD International Conference on Management of Data (SIGMOD)*, San Francisco, USA, June 27, 2016

HONORS AND AWARDS

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|---|----------|
| • Top Minds Program , Huawei, China | 2021 |
| • Selected for Rising Stars in EECS Workshop , UIUC, USA | 2019 |
| – Winner of the Research Pitch Competition (ECE Track) | |
| • ECE Woman of the Month , Cornell University, USA | Mar 2018 |
| • Cornell Graduate Fellowship , Cornell University, USA | 2015 |
| • Outstanding Graduate (top 1%), Shanghai Jiao Tong University, China | 2015 |
| • Outstanding Graduate in Zhiyuan College (top 12), Shanghai Jiao Tong University, China | 2015 |
| • Kai Yuan Scholarship (top 20 in Zhiyuan College), Shanghai Jiao Tong University, China | 2013 |
| • ACM-ICPC (International Collegiate Programming Contest) Asia Regional | |
| – 4th place in Hsinchu Site | 2011 |
| – Gold medal in Beijing Site | 2011 |
| • Silver medal (ranked the 64th) in National Olympiad in Informatics, China | 2010 |

TEACHING EXPERIENCES

Cornell University, USA

- **Teaching Assistant** of Advanced Computer Architecture (CS5420/ECE5750) Fall 2020, Fall 2019
- **Guest Lecturer** of Competition Programming and Problem Solving Seminar (CS5199) Sept 2019
- **Head Teaching Assistant** of Computer Systems Programming (ECE2400) Spring 2017
- **Teaching Assistant** of Computer Architecture (CS4420/ECE4750) Fall 2016

Shanghai Jiao Tong University, China

- **Teaching Assistant** of C++ Programming (CS110) Fall 2012